

Incident Management System (IMS)

Project Overview

The Incident Management System (IMS) is a web-based application designed to manage, track, and resolve incidents efficiently. This project contains both frontend and backend services along with Docker support for easy deployment.

Repository Structure

```
Engineering-Challenge--IMS-  
|  
├── backend/           # Backend application source code  
├── frontend/         # Frontend application source code  
├── README.md          # Project documentation  
└── docker-compose.yml # Docker configuration file
```

Features

- Incident creation and management
 - Incident status tracking
 - Frontend and backend integration
 - Dockerized deployment
 - REST API support
 - Scalable architecture
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Technologies Used

Frontend

- HTML
- CSS
- JavaScript
- React (if used)

Backend

- Node.js / Express (if used)
- REST API
- Database integration

DevOps & Deployment

- Docker
 - Docker Compose
 - GitHub
-

Prerequisites

Before running the project, ensure the following tools are installed:

- Git
 - Docker
 - Docker Compose
 - Node.js
 - npm
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Clone the Repository

```
git clone https://github.com/Madhava9800/Engineering-Challenge--IMS-.git
cd Engineering-Challenge--IMS-
```

Running the Application

Using Docker

Run the following command:

```
docker-compose up --build
```

This command will:

- Build frontend and backend containers
- Start all services
- Run the application locally

Backend Setup

```
cd backend
npm install
npm start
```

Frontend Setup

```
cd frontend
npm install
npm start
```

API Endpoints

Method	Endpoint	Description
GET	/incidents	Get all incidents
POST	/incidents	Create new incident
PUT	/incidents/:id	Update incident
DELETE	/incidents/:id	Delete incident

Docker Commands

Start Containers

```
docker-compose up
```

Stop Containers

```
docker-compose down
```

Rebuild Containers

```
docker-compose up --build
```

GitHub Repository

Repository Link:

```
https://github.com/Madhava9800/Engineering-Challenge--IMS-
```

Future Improvements

- User authentication
 - Role-based access control
 - Incident analytics dashboard
 - Notification system
 - Kubernetes deployment
 - CI/CD pipeline integration
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Author

Madhava Akula

License

This project is developed for learning and engineering challenge purposes.